Statistical Analysis of Radioactive Decay: A Comparison of Poisson and Gaussian Distributions

Group 2
Aneleh Du Plessis (46858504)
Andrew Nare (31506534)
Christoffel Schoeman (38655349)
Kaylyn Smit (41578708)

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Contents

1 Summary

In science, uncertainty is not seen as a limitation but rather as an essential feature that provides insight into the nature of certain physical events. Statistical variations serve as a mathematical framework for assessing uncertainty. In this experiment, radioactive decay shows statistical variations in the experimental measurements. To demonstrate these variations, the data will be processed using both the Poisson method and the Gaussian method.

Using a Geiger counter, radiation events from a cobalt source are recorded, firstly at a far distance, then at a close distance. Calculations are then made to determine the frequency distribution, average, and standard deviation. Using that information, tables are drawn for the theoretical values of the Gaussian and Poisson distributions. After careful assessment, a determination is made as to whether the Poisson or Gaussian distribution is a better fit for the data. [1]

2 Introduction

This experiment consists of observing the radioactive decay of cobalt as a source. A Geiger counter is used to count the number of radiation particles per specific time interval, which is called a count rate. The count rate is measured first at a further distance and then at a close distance. The data is then statistically interpreted.

In this experiment, three key probability distributions are used:

- The Binomial distribution:

  For N independent events, with probability p, the binomial distribution is given by

  $$f_{(N,p)}(n_i) = \binom{N}{n_i} p^{n_i} q^{N-n_i}$$

  , where there is a probability of $q = 1 - p$ to be unsuccessful.

- The Poisson distribution:

  This distribution is given by

  $$P(n_i) = \frac{\overline{n}^{n_i} e^{-\overline{n}}}{n_i!}, \quad n_i = 0, 1, 2, \ldots, N.$$

  , where for a fixed interval, it gives the probability of n events occurring.

- The Gaussian distribution:

  This shows a distribution of random errors and is given by

  $$G(\Delta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\Delta^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(n-\overline{n})^2}{2\sigma^2}}$$

The objective of this experiment can be divided into two parts. Firstly, the event of radioactive decay is to be analyzed through the calculation of key statistical parameters (such as the frequency distribution, mean, and standard deviation). Through these calculations it is then determined which different statistical models apply to the data. The second important part is where the probability distributions are validated. This is done by performing the chi-squared tests as well as assessing the transition between the different distribution models.

The primary hypothesis for this experiment is that the events of radioactive decay will follow a Poisson distribution for the farther observation and a Gaussian distribution for the close observation. The reasoning would be that for the close observation, more counts of radiation will be measured. When there are smaller counts measured ($\lambda < 20$), the result will be an asymmetric distribution. For the larger count measurements,($\lambda \geq 20$), the result will be a symmetrical bell-curve distribution.

3 Experimental Method

Firstly, the experiment is set up. The Geiger counter is placed on a flat surface and connected to a computer that has a data-logging system in place. The Geiger counter is calibrated to ensure accurate measurements. The cobalt source is then placed far from the counter, with the Geiger counter facing the source. The count rate is then measured, and the data is stored. Then, the cobalt source is moved to a closer distance, and the count rate is measured again, with the data stored as well. Using the data from when the source was further away, the frequency distribution is calculated using the formula $\frac{f(n_i)}{N}$, where

$$f(n_i) = \frac{\overline{n}^{n_i}}{n_i!}[(1-p)^{\frac{1}{p}}]^{\overline{n}}$$

The mean ($\overline{n}$) and standard deviation (σ) is also calculated using formulas:

$$\overline{n} = \sum_{i=1}^{N} n_i f(n_i)$$

$$\sigma = [\sum_{i=1}^{N}(n_i - \overline{n})^2 f(n_i)]^{\frac{1}{2}}$$

Next, the theoretical distributions of Poisson and Gaussian are calculated using the previously stated equations. This process is then repeated using the data from when the cobalt source was close to the counter. After that, the goodness-of-fit analysis is done using the chi-squared tests. The data is binned into classes such that each class has at least five occurrences, but the total amount of classes is less than thirty. The formula for chi-square is given by

$$\chi^2 = \sum_{j=1}^{M} \frac{(m_j - t_j)^2}{t_j}$$

Looking at the chi-squared results, the degrees of freedom are then determined using v.

- For Poisson: $v = M - 2$
- For Gaussian: $v = M - 3$

Lastly, the hypothesis is either accepted or rejected according to how well the data satisfies the hypothesis.

4 Results

4.1 Measurements at a Far Distance

4.1.1 Frequency distribution

Minimum	Max	n_i	$f(n_i)$	$F(n_i)$
0	7	0	38	0.126667
		1	62	0.206667
		2	89	0.296667
		3	63	0.21
		4	33	0.11
		5	9	0.03
		6	4	0.013333
		7	2	0.006667
Total			300	1

Table 1: Frequency Distribution Table

4.1.2 Average

n_i	$F(n_i)$	$n_i \cdot F(n_i)$
0	0.126667	0
1	0.206667	0.206667
2	0.296667	0.593333
3	0.21	0.63
4	0.11	0.44
5	0.03	0.15
6	0.013333	0.08
7	0.006667	0.046667
Average		**2.146667**

Table 2: Relative Frequency Calculation

4.1.3 Standard Deviation

n_i	$F(n_i)$	$(n_i - \bar{n})^2 \cdot F(n_i)$
0	0.126667	0.583702519
1	0.206667	0.271734519
2	0.296667	0.00638163
3	0.21	0.152917333
4	0.11	0.377832889
5	0.03	0.244245333
6	0.013333	0.197975704
7	0.006667	0.157032296
Standard Deviation		**1.991822222**

Table 3: Standard Deviation Calculation

4.1.4 Poisson Theoretical Values

n_i	$P(n_i)$	$NP(n_i)$
0	0.116873086	35.06192585
1	0.250887558	75.26626748
2	0.269285979	80.78579376
3	0.192689078	57.80672354
4	0.103409805	31.02294163
5	0.044397276	13.31918294
6	0.015884359	4.765307674
7	0.004871203	1.46136102
Total	**0.998298346**	**299.4895039**

Table 4: Poisson Distribution Table

4.1.5 Gaussian Theoretical Values

n_i	$G(n_i)$	$NG(n_i)$
0	0.112056699	33.61700964
1	0.169704541	50.9113622
2	0.19974785	59.9243551
3	0.182727444	54.81823313
4	0.129914748	38.9744245
5	0.071787057	21.53611699
6	0.030829526	9.248857866
7	0.010290123	3.087036909
Total	**0.907057988**	**272.1173963**

Table 5: Gaussian Distribution Table

4.1.6 Classes for Chi-Values

Measured Values				**Poisson Distribution**			**Gaussian Distribution**		
Classes	j	m_j	Nm_j	t_j	t_j Values	Nt_j	t_j	t_j Values	Nt_j
Group 0	1	0.1267	38	1	0.1169	35.06	1	0.1121	33.62
Group 1	2	0.2067	62	2	0.2509	75.27	2	0.1697	50.91
Group 2	3	0.2967	89	3	0.2693	80.79	3	0.1997	59.92
Group 3	4	0.2100	63	4	0.1927	57.81	4	0.1827	54.82
Group 4	5	0.1100	33	5	0.1034	31.02	5	0.1299	38.97
Group 5	6	0.0300	9	6	0.0444	13.32	6	0.0718	21.54
Group 6 and 7	7	0.0200	6	7	0.0208	6.23	7	0.0411	12.34
Total		1.0000	300		0.9983	299.49		0.9071	272.12

Table 6: Classes for Chi-Values

4.1.7 Chi Calculations

Chi-squared Poisson		Chi-squared Gaussian	
j	Chi-component	j	Chi-component
1	0.00082067	1	0.00190485
2	0.007794278	2	0.008050455
3	0.002784037	3	0.047025572
4	0.001555189	4	0.004070501
5	0.000419986	5	0.00305275
6	0.00466879	6	0.024324136
7	0.0000275046	7	0.010847359
Total	0.018070454	Total	0.099275622
V	5	V	4
χ^2/V	0.003614091	χ^2/V	0.024818906

Table 7: Chi-squared Poisson and Gaussian Components

4.2 Measurements at a Near Distance

4.2.1 Frequency distribution

Table 8: Table of n_i, $f(n_i)$, and $F(n_i)$ values (Min = 31, Max = 71)

n_i	$f(n_i)$	$F(n_i)$
31	1	0.0033
32	0	0
33	2	0.0067
34	0	0
35	3	0.0100
36	0	0
37	2	0.0067
38	5	0.0167
39	4	0.0133
40	5	0.0167
41	8	0.0267
42	11	0.0367
43	14	0.0467
44	12	0.0400
45	13	0.0433
46	13	0.0433
47	15	0.0500
48	9	0.0300
49	26	0.0867
50	16	0.0533
51	22	0.0733
52	12	0.0400
53	15	0.0500
54	18	0.0600
55	7	0.0233
56	16	0.0533
57	3	0.0100
58	8	0.0267
59	12	0.0400
60	5	0.0167
61	8	0.0267
62	2	0.0067
63	2	0.0067
64	4	0.0133
65	1	0.0033
66	2	0.0067
67	1	0.0033
68	1	0.0033
69	0	0
70	1	0.0033
71	1	0.0033
Total	300	1

4.2.2 Average

n_i	$F(n_i)$	$n_i \cdot F(n_i)$
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	0	0
10	0	0
11	0	0
12	0	0
13	0	0
14	0	0
15	0	0
16	0	0
17	0	0
18	0	0
19	0	0
20	0	0
21	0	0
22	0	0
23	0	0
24	0	0
25	0	0
26	0	0
27	0	0
28	0	0
29	0	0
30	0	0
31	0.003333333	0.103333333
32	0	0
33	0.006666667	0.22
34	0	0
35	0.01	0.35
36	0	0
37	0.006666667	0.246666667

n_i	$F(n_i)$	$n_i * F(n_i)$
38	0.016666667	0.633333333
39	0.013333333	0.52
40	0.016666667	0.666666667
41	0.026666667	1.093333333
42	0.036666667	1.54
43	0.046666667	2.006666667
44	0.04	1.76
45	0.043333333	1.95
46	0.043333333	1.993333333
47	0.05	2.35
48	0.03	1.44
49	0.086666667	4.246666667
50	0.053333333	2.666666667
51	0.073333333	3.74
52	0.04	2.08
53	0.05	2.65
54	0.06	3.24
55	0.023333333	1.283333333
56	0.053333333	2.986666667
57	0.01	0.57
58	0.026666667	1.546666667
59	0.04	2.36
60	0.016666667	1
61	0.026666667	1.626666667
62	0.006666667	0.413333333
63	0.006666667	0.42
64	0.013333333	0.853333333
65	0.003333333	0.216666667
66	0.006666667	0.44
67	0.003333333	0.223333333
68	0.003333333	0.226666667
69	0	0
70	0.003333333	0.233333333
71	0.003333333	0.236666667

Table 9: Relative Frequency Calculation

4.2.3 Standard Deviation

n_i	$F(n_i)$	**Standard Deviation Components**
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	0	0
10	0	0
11	0	0
12	0	0
13	0	0
14	0	0
15	0	0
16	0	0
17	0	0
18	0	0
19	0	0
20	0	0
21	0	0
22	0	0
23	0	0
24	0	0
25	0	0
26	0	0
27	0	0
28	0	0
29	0	0
30	0	0
31	0.0033	1.2203
32	0	0
33	0.0067	1.9570
34	0	0
35	0.0100	2.2902
36	0	0
37	0.0067	1.1499
38	0.0167	2.4536

n_i	$F(n_i)$	**Standard Deviation Components**
39	0.0133	1.6527
40	0.0167	1.7114
41	0.0267	2.2245
42	0.0367	2.4255
43	0.0467	2.3746
44	0.0400	1.5047
45	0.0433	1.1419
46	0.0433	0.7403
47	0.0500	0.4909
48	0.0300	0.1365
49	0.0867	0.1113
50	0.0533	0.0009
51	0.0733	0.0551
52	0.0400	0.1394
53	0.0500	0.4109
54	0.0600	0.8971
55	0.0233	0.5526
56	0.0533	1.8356
57	0.0100	0.4715
58	0.0267	1.6503
59	0.0400	3.1447
60	0.0167	1.6225
61	0.0267	3.1489
62	0.0067	0.9388
63	0.0067	1.1037
64	0.0133	2.5638
65	0.0033	0.7367
66	0.0067	1.6783
67	0.0033	0.9483
68	0.0033	1.0641
69	0	0
70	0.0033	1.3156
71	0.0033	1.4514

Table 10: Table of n_i, $F(n_i)$, and standard deviation components. The standard deviation is 7.0225.

4.2.4 Poisson Theoretical Values

n_i	$P(n_i)$	$NP(n_i)$
1	8.46274E-21	2.53882E-18
2	2.12132E-19	6.36397E-17
3	3.54497E-18	1.06349E-15
4	4.44302E-17	1.33291E-14
5	4.45487E-16	1.33646E-13
6	3.72229E-15	1.11669E-12
7	2.66586E-14	7.99759E-12
8	1.67061E-13	5.01182E-11
9	9.30589E-13	2.79177E-10
10	4.66535E-12	1.39961E-09
11	2.12627E-11	6.3788E-09
12	8.88307E-11	2.66492E-08
13	3.42567E-10	1.0277E-07
14	1.22672E-09	3.68015E-07
15	4.09996E-09	1.22999E-06
16	1.28465E-08	3.85396E-06
17	3.78846E-08	1.13654E-05
18	1.05516E-07	3.16547E-05
19	2.78413E-07	8.35239E-05
20	6.97888E-07	0.000209366
21	1.66607E-06	0.00049982
22	3.79661E-06	0.001138984
23	8.27551E-06	0.002482653
24	1.72866E-05	0.005185983
25	3.46654E-05	0.010399617
26	6.68419E-05	0.020052582
27	0.000124111	0.037233411
28	0.000222218	0.066665491
29	0.000384156	0.115246933
30	0.000641968	0.192590302
31	0.001038192	0.311457657
32	0.0016265	0.487950005
33	0.002470964	0.741289212
34	0.003643458	1.093037484
35	0.005218817	1.565645031
36	0.007267681	2.180304223
37	0.009847373	2.954212046
38	0.012991614	3.897484178

n_i	$P(n_i)$	$NP(n_i)$
39	0.01670032	5.010095988
40	0.020931054	6.27931613
41	0.025593727	7.678118033
42	0.030549952	9.16498559
43	0.035617905	10.68537144
44	0.040582798	12.17483937
45	0.045212213	13.56366388
46	0.049274727	14.78241805
47	0.052559674	15.7679021
48	0.054895623	16.4686868
49	0.05616528	16.849584
50	0.056315017	16.89450499
51	0.055357993	16.60739779
52	0.053370747	16.01122414
53	0.050483994	15.14519818
54	0.046869059	14.06071785
55	0.042721829	12.81654884
56	0.038246184	11.47385514
57	0.033638726	10.09161793
58	0.029076213	8.722863948
59	0.024706551	7.411965342
60	0.020643682	6.193104701
61	0.016966163	5.089848786
62	0.013718866	4.115659937
63	0.010917017	3.27510499
64	0.008551657	2.565497203
65	0.006595735	1.978720629
66	0.005010091	1.503027196
67	0.003748841	1.124652438
68	0.00276385	0.829154971
69	0.002008129	0.602438767
70	0.001438202	0.431460621
71	0.001015519	0.30465556
Total	0.997858061	299.3574184

Table 11: Table of n_i, $P(n_i)$, and $NP(n_i)$.

4.2.5 Gaussian Theoretical Values

n_i	$G(n_i)$	$NG(n_i)$
1	1.33256E-12	3.99769E-10
2	3.57251E-12	1.07175E-09
3	9.38542E-12	2.81563E-09
4	2.41617E-11	7.2485E-09
5	6.09528E-11	1.82859E-08
6	1.5068E-10	4.52039E-08
7	3.65013E-10	1.09504E-07
8	8.66475E-10	2.59942E-07
9	2.01557E-09	6.0467E-07
10	4.59443E-09	1.37833E-06
11	1.02627E-08	3.0788E-06
12	2.24637E-08	6.73912E-06
13	4.81834E-08	1.4455E-05
14	1.01276E-07	3.03828E-05
15	2.08597E-07	6.25792E-05
16	4.21022E-07	0.000126307
17	8.32712E-07	0.000249813
18	1.6139E-06	0.000484171
19	3.06517E-06	0.000919551
20	5.7046E-06	0.001711379
21	1.04037E-05	0.003121117
22	1.85929E-05	0.005577858
23	3.2561E-05	0.009768288
24	5.58781E-05	0.016763436
25	9.3968E-05	0.028190394
26	0.00015485	0.046455039
27	0.000250056	0.075016719
28	0.00039569	0.118707146
29	0.000613575	0.184072633
30	0.000932339	0.279701724
31	0.001388268	0.416480451
32	0.002025659	0.607697795
33	0.002896363	0.868908927
34	0.004058197	1.217459205
35	0.005571946	1.671583734
36	0.00749677	2.249030863
37	0.009884051	2.965215447
38	0.012769956	3.830986773

n_i	$G(n_i)$	$NG(n_i)$
39	0.016167295	4.850188429
40	0.020057596	6.017278739
41	0.024384506	7.315351665
42	0.029049763	8.714928875
43	0.033912889	10.17386676
44	0.038795428	11.63862845
45	0.043490046	13.04701383
46	0.047774126	14.33223767
47	0.051426763	15.42802876
48	0.054247432	16.27422958
49	0.056074156	16.82224667
50	0.056798892	17.03966747
51	0.056378113	16.91343402
52	0.054837138	16.45114128
53	0.052267602	15.68028061
54	0.048818444	14.64553324
55	0.044681614	13.4044842
56	0.040074428	12.02232837
57	0.035220812	10.56624372
58	0.030333671	9.100101206
59	0.025600245	7.680073576
60	0.021171754	6.351526274
61	0.01715786	5.147358031
62	0.01362583	4.087748948
63	0.010603674	3.181102139
64	0.008086178	2.425853512
65	0.0060426	1.812779954
66	0.004424843	1.327453046
67	0.00317516	0.952547862
68	0.002232681	0.669804413
69	0.001538443	0.461533004
70	0.001038795	0.311638483
71	0.00068734	0.206202041
Total	0.998837278	299.6511833

Table 12: Table of n_i, $G(n_i)$, and $NG(n_i)$.

4.2.6 Classes for Chi-Values

Measured Values				Poisson Distribution			Gaussian Distribution		
Classes	j	m_j	Nm_j	t_j	t_j Values	Nt_j	t_j	t_j Values	Nt_j
Group 1 to 30	1	0.4518	0.0015	1	0.7709	0.0026	1	0.4518	0.0015
Group 31 to 37	2	9.3339	0.0311	2	9.9964	0.0333	2	9.9964	0.0333
Group 38 to 61	3	272.8413	0.9095	3	273.4472	0.8987	3	272.8413	0.9095
Group 62 to 71	4	16.7304	0.0558	4	15.4367	0.0515	4	16.7304	0.0558
Total		299.3574	0.9979		299.6512	0.9861		299.3574	0.9979

Table 13: Classes for Chi-Values

4.2.7 Chi Calculations

Poisson-Chi Calculation		Gaussian-Chi Calculation	
j	$(m_j - t_j)^2/t_j$	j	$(m_j - t_j)^2/t_j$
1	0.001506124	1	0.002569951
2	0.000635418	2	0.001328988
3	0.000211291	3	0.000674056
4	0.000596557	4	4.11736E-05
Total	0.002949391	**Total**	0.004614168
V	2	V	1
X/v	0.001474695	X/v	0.004614168

Table 14: Chi-squared Poisson and Gaussian Components

5 Conclusion

This experiment investigated statistical variations in radioactive decay by analysing count rates using a Geiger counter at various distances from a cobalt source. The data was analysed using Poisson and Gaussian distributions, with essential statistical metrics such as mean, standard deviation, and frequency distributions calculated.

The results support the initial hypothesis: at a greater distance, the count rate followed a Poisson distribution due to lower event counts; however, at a closer distance, the count rate transitioned to a Gaussian distribution, as expected for a higher number of recorded events. These findings were supported by chi-squared tests, which validated the shift between distributions.

This experiment shows how statistical methods can be used to describe radioactive decay, with Poisson statistics better suited to low-count cases and the Gaussian approximation becoming more valid as event counts increase. These results are consistent with theoretical expectations and highlight the relevance of statistical analysis in experimental physics.

6 Assignment Questions

6.1 Assignment 1

Show that $\overline{\triangle} = 0$

The mean deviation is calculated using the following formula:

$$\overline{\triangle} = \frac{1}{N}\sum_{i=0}^{N}(n_i - \overline{n})$$

Where:

- n_i represents each value in a dataset.
- n is the mean (average) of all values.
- N is the total number of values.

The mean is given by:

$$\overline{n} = \frac{1}{N}\sum_{i=0}^{N} n_i$$

Substituting n into the Mean Deviation Formula:

$$\overline{\triangle} = \frac{1}{N}\sum_{i=0}^{N} n_i - \frac{1}{N}\sum_{i=0}^{N}\overline{n}$$

Simplifying the Second Term, since n is a constant, the sum of over N terms is:

$$\frac{1}{N}\sum_{i=0}^{N}\overline{n} = \frac{N\overline{n}}{N} = \overline{n}$$

Simplifying:

$$\overline{\triangle} = \overline{n} - \overline{n} = 0$$

Thus, the sum of deviations from the mean is zero.

6.2 Assignment 2

Consider two dice and one coin being tossed and thrown. What is the probability of finding the number 7, plus one head?

We first determined the probability of obtaining a sum of 7 when rolling two dice.

The possible outcomes that sum to 7 are: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)

There are 6 favourable outcomes, and since each die has 6 faces, the total number of possible outcomes when rolling two dice is: $6 \times 6 = 36$ Therefore, the probability of rolling a sum of 7 is: $P_{sum}(7) = \frac{6}{36} = \frac{1}{6}$

Determining the probability of getting one head when flipping a coin, the probability of getting heads is: $P(H) = \frac{1}{2}$

Since the dice roll and coin flip are independent events, we multiply their probabilities: $P(7and1head) = P(7) \times P(H) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12} \approx 0.0833$ Thus, the final probability is: 8.33%

6.3 Assignment 6

Plot the following binomial distributions (histograms) on one plot: $f_{5,\frac{3}{4}}, f_{20,\frac{1}{2}} f_{20,\frac{3}{10}} f_{20,\frac{1}{20}}$‘. What can you infer?

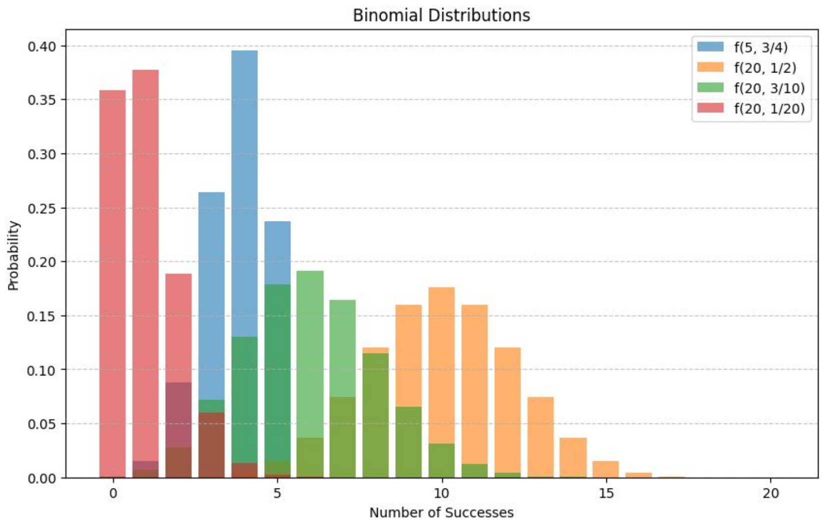

Figure 1: Stacked Histogram

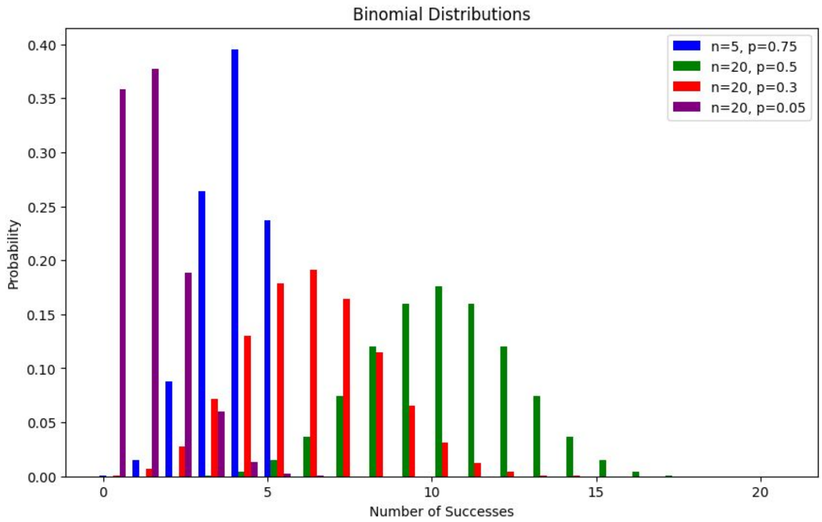

Figure 2: Side-by-side Histogram

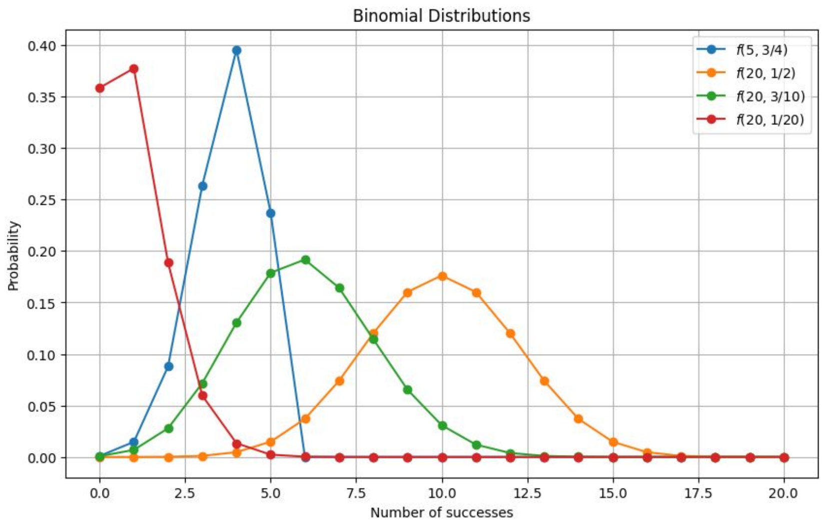

Figure 3: Line chart

Looking at the line chart for a clearer demonstration, the following observations can be made:

The $f_{5,\frac{3}{4}}$ distribution is skewed to the right since most result in a high number of successes. The $f_{20,\frac{1}{2}}$ distribution is symmetric, resembling a normal curve, since successes and failures are equally likely. The $f_{20,\frac{3}{10}}$ distribution shifts to the left, indicating that lower numbers of successes occur more frequently. The $f_{20,\frac{1}{20}}$ distribution is highly skewed to the left, as successes are rare.

7 Group Contribution

- Aneleh Du Plessis (46858504) - Writing: Introduction, Method and Summary
- Andrew Nare (31506534) - Assignments & Conclusion Writing
- Christoffel Schoeman (38655349) - Results Analysis & Interpretation.
- Kaylyn Smit (41578708) - LaTeX report Compilation & Formatting.

References

[1] Physics third-year experiments (2025 s1), 01 2025.